

Pedagogy Judges: A Multi-Model LLM Ensemble for Scoring Teaching Quality

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Research Question

Large language models are increasingly used to evaluate teaching quality in AI-tutoring transcripts, but a single LLM's judgment is an unreliable ground truth — it reflects one model's biases and blind spots, not a validated measure. This project asks: *can independent scores from multiple, architecturally different LLMs be combined to produce a more trustworthy pedagogy assessment, and where do such models agree or disagree?*

Methodology

A 6-factor pedagogy rubric (1–5 scale per factor) was built on established teaching-quality research: **PCK** (Shulman), **Feedback** (Hattie's Visible Learning), **Modeling** and **Coaching** (Cognitive Apprenticeship), **Reflection** (student articulation of reasoning), and **Engagement** (the ICAP framework's Passive→Active→Constructive→Interactive scale). The overall Pedagogy Score is $(\text{sum of 6 factor scores} / 30) \times 100$.

Three judges — OpenAI (gpt-5-mini), Groq (gpt-oss-120b), and Google Gemini — each receive the identical system prompt and rubric and independently score the same lesson transcript, returning structured JSON with a score, confidence, evidence, reasoning, strengths, weaknesses, and suggestions per factor. Because all three see the same instructions, any disagreement between them reflects genuine differences in model judgment rather than prompt variation. Per-factor scores are aggregated by mean, with the standard deviation across the three judges reported as an inter-rater agreement signal.

Results

On a sample transcript (a binary-search lesson with worked examples, paired practice, and peer explanation), the ensemble produced:

83.4 / 100

Aggregated Pedagogy Score

"Very Good pedagogy" — consistently implemented practices that positively support student learning.

Factor	OpenAI	Groq	Gemini	Mean	Agreement (stdev)
PCK	4	4	4	4.00	0.00
Feedback	4	4	4	4.00	0.00
Modeling	5	5	4	4.67	0.47
Coaching	4	4	4	4.00	0.00
Reflection	3	4	4	3.67	0.47
Engagement	5	5	4	4.67	0.47

The three judges reached **exact unanimous agreement** on half the rubric (PCK, Feedback, Coaching). Disagreement was mild and consistent everywhere it occurred: on Modeling, Reflection, and Engagement, two judges agreed and one differed by exactly one point — no judge diverged by more than a single point on any factor. This is the intended signal of the multi-judge design: it surfaces exactly where an evaluation is contestable instead of presenting a single confident-looking number.

Limitations & Next Steps

- No human ground truth yet.** Judges agreeing with each other shows consistency, not correctness — the next step is comparing judge scores to real human expert ratings.
 - Aggregation is currently a plain mean.** Each judge already outputs a confidence score per factor that the aggregation step does not yet use.
- Single-transcript demonstration.** A larger paired teacher-student dataset (278 experiment runs across multiple student/tutor model combinations) is already collected and ready for batch evaluation.
 - Per-episode scoring** (scoring one question within a multi-question session) is not yet implemented.

Code and full results: github.com/aryankumar1124/pedagogy_judges